Friction changes when slipping begins

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Static friction / ES: fricción estática / PT: atrito estático

Adjusts to prevent slip, up to a limit.

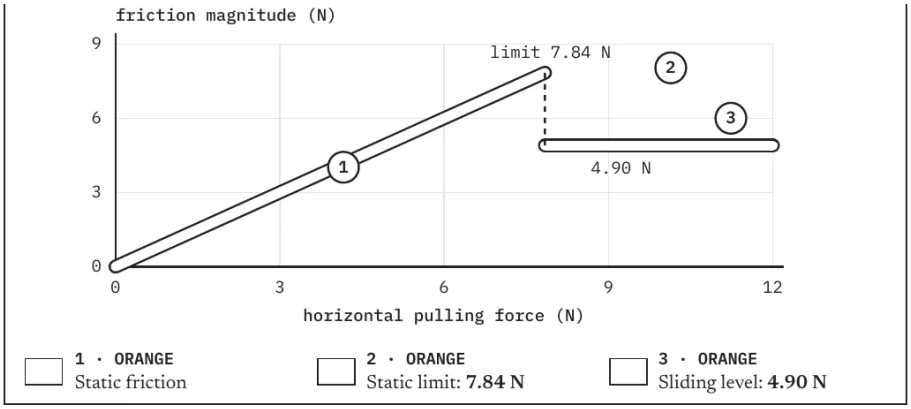
Also on your note page: Limit, Kinetic friction, Normal force.

B READ THE RELATIONSHIP

Static friction adjusts up to a limit; sliding friction acts during slip

As you read, find: at rest · just before slip · sliding.

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

For this box, compare a 6 N pull while at rest with a 10 N pull after sliding starts. Use g = 9.8 m/s² ; pulls are horizontal.

Given N = 2(9.8) = 19.6 N μ s = 0.40 μ k = 0.25

Rule Check μ s N . Sliding: use μ k N , then net force.

Work f s ,max = 7.84 N at rest: f s = 6 N a = (10 – 4.9)/2

Answer 0; 2.55 m/s² At rest, then sliding.

E SAVE FOR THE FINAL MEETING

Surface friction is not a force that continually brakes XJ in empty space. Gravitational forces can still change its velocity, and atmospheric drag would matter in an atmosphere.

Read the physics: Static friction

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Push gently on a crate. If it stays still, static friction can balance the pull. A harder push needs more static friction, up to a maximum.

02 HOW IT WORKS

After slipping begins, use kinetic friction. In this model its size is the kinetic coefficient times the normal force. It opposes relative slipping at the contact.

03 WHAT TO CHECK

On a force trace, find the peak just before slip. Compare it with the steady-speed pull while the surfaces slide. Do not use the maximum static force for every pull.

04 TRY THE EXAMPLE FROM YOUR NOTES

For this box, compare a 6 N pull while at rest with a 10 N pull after sliding starts. Use g = 9.8 m/s² ; pulls are horizontal. 0; 2.55 m/s² At rest, then sliding.

Model check: Static friction adjusts up to a limit; sliding friction acts during slip.

05 CONNECT TO XJ

Surface friction is not a force that continually brakes XJ in empty space. Gravitational forces can still change its velocity, and atmospheric drag would matter in an atmosphere.

SAY IT BEFORE YOU WRITE

When do you switch from static to kinetic friction? Before sliding, friction ___. During sliding, we use ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Static friction

ES: fricción estática / PT: atrito estático

Meaning in my words	How can you tell?
How would you explain static friction to a friend? In my words, this means ___.	You push a crate gently and it stays still. What force balances your sideways push? I can tell because ___.
Example or sketch	**Non-example and why**
A stationary crate on a level floor is pushed right with 3 N below the sliding threshold. Draw friction. My example shows ___.	The crate is already sliding across the floor. Would the friction at that contact still be static? I would change ___ because ___.

Limit

ES: límite / PT: limite

Meaning in my words	How can you tell?
How would you explain limit to a friend? In my words, this means ___.	The floor can supply up to 6 N. How much friction balances a gentle 2 N horizontal push? I can tell because ___.
Example or sketch	**Non-example and why**
With a 6 N limit and no other horizontal forces, what do you predict for a 7 N push? My example shows ___.	A friend writes 6 N for every trial, even with no sideways push. What does 6 N mean? I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Kinetic friction

ES: fricción cinética / PT: atrito cinético

Meaning in my words	How can you tell?
How would you explain kinetic friction to a friend? In my words, this means ___.	A crate slides right across a stationary floor. Which way does kinetic friction act on the crate? I can tell because ___.
Example or sketch	**Non-example and why**
If the coefficient is 0.2 and the normal force is 10 N, what arrow would you draw? My example shows ___.	A crate stays put while you push gently. Should you use the kinetic-friction rule? I would change ___ because ___.

Normal force

ES: fuerza normal / PT: força normal

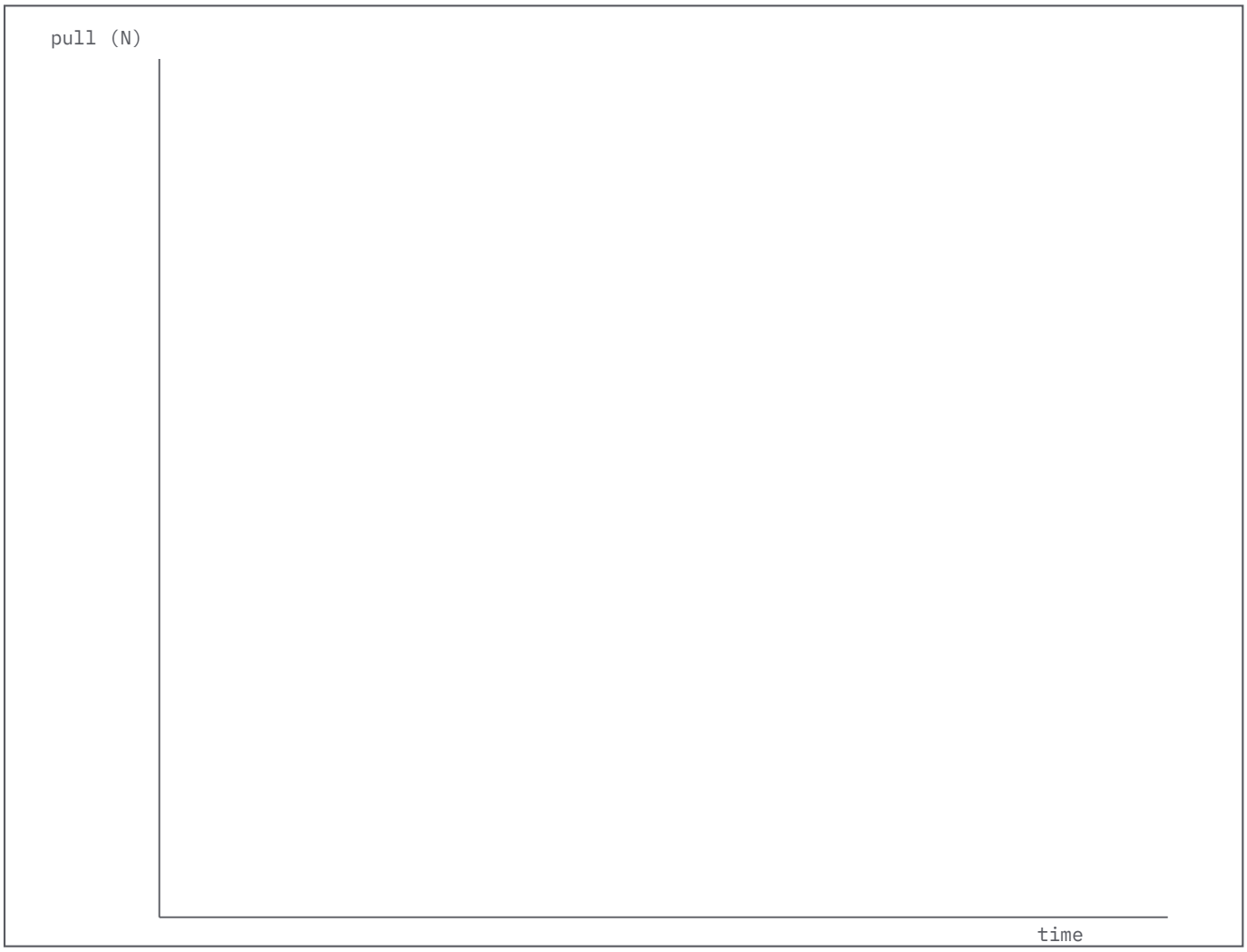
Meaning in my words	How can you tell?
How would you explain normal force to a friend? In my words, this means ___.	A crate rests on a ramp. Which way does the ramp push the crate? I can tell because ___.
Example or sketch	**Non-example and why**
How would you draw the normal force on a book on a level table? My example shows ___.	Someone draws the normal force along the ramp. What would you rotate? I would change ___ because ___.

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Sketch a force trace. Label the peak just before slip and the steady-speed pull while the surfaces slide.

Labels you can use: at rest · just before slip · sliding.



Tell a partner

Before sliding, friction ___. During sliding, we use ___.

Cut from the table to space

17

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Use contact surfaces to decide when friction acts.

CHOOSE ONE JOB TOGETHER

EFFECTS ENGINEER ☐

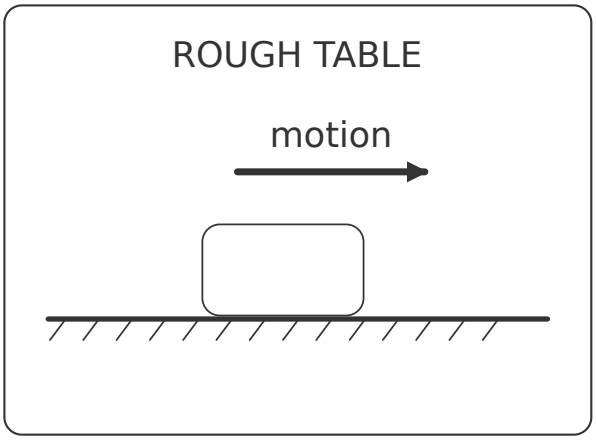
Set friction for each scene. Mark the touching surfaces and explain what could change XJ's motion in empty space.

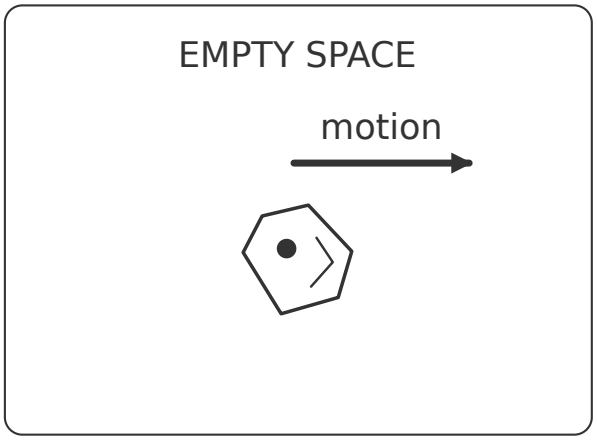
SCIENCE BROADCASTER ☐

Correct the caption for the two shots. Point to the surfaces and say what could still bend XJ's path.

Word help: contact / friction / drag

A show cuts from a block sliding on a rough table to XJ coasting in empty space. The proposed caption says “Both will stop from surface friction.”





01 OUR FIRST CALL

What should the two-scene caption say? Explain using the contact surfaces.

In this scene, ___ touches ___, so ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

What changed in the new shot? Update the force explanation for XJ.

Now we would say ___ because ___.

Switch speakers. How would entering an atmosphere change the forces you include? Explain where the added force comes from.

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks